The Zero-Privilege Pivot

Deploying a live dashboard on a restricted Ubuntu container without Docker or Root access

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The Mission

Goal

Interactive web vault for CRISPR selection process

Constraints

- No Root access
- No Docker Engine
- Old Node.js environment

Reality

Survival mode engineering on shared infrastructure

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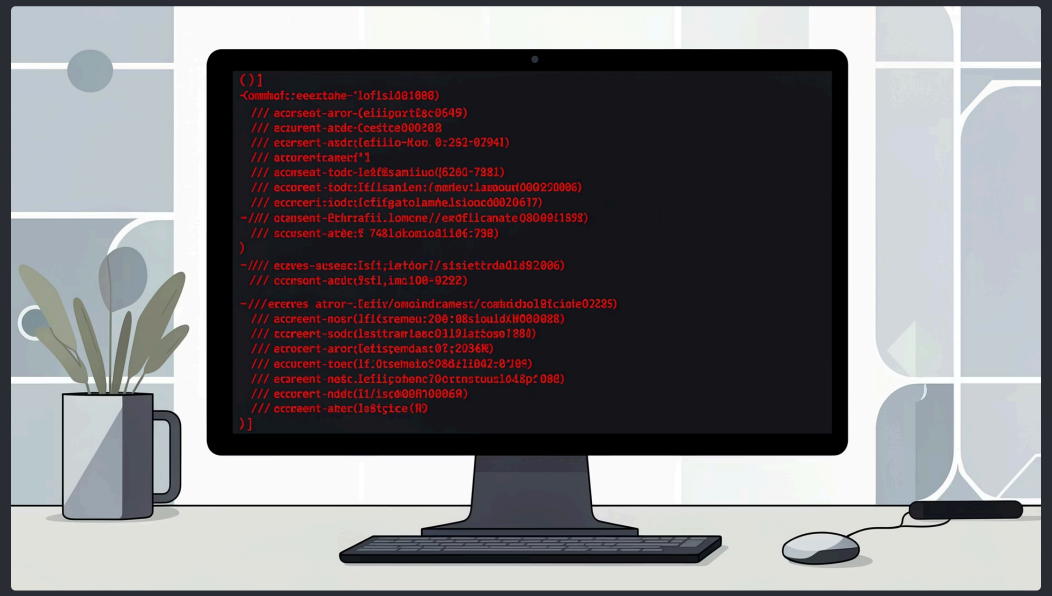
What I Broke First



The Mistake

Trying "modern" tools like Coolify and Docker on a restricted environment

The Result

A wall of red text. Automated scripts failed spectacularly.

> **The Lesson:** You can't build a skyscraper on a foundation you don't own. The "standard" way is often the "lazy" way.

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Architecture Pivot

Backend

Express.js v4.17.1 for legacy compatibility

Filesystem

Native fs module reading markdown in real-time

Network

Bound to 0.0.0.0:8000 for internal bridging

Minimalist Node.js process. No container manager, no deployment automation—just raw process control.

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Software Compaibillity Digram

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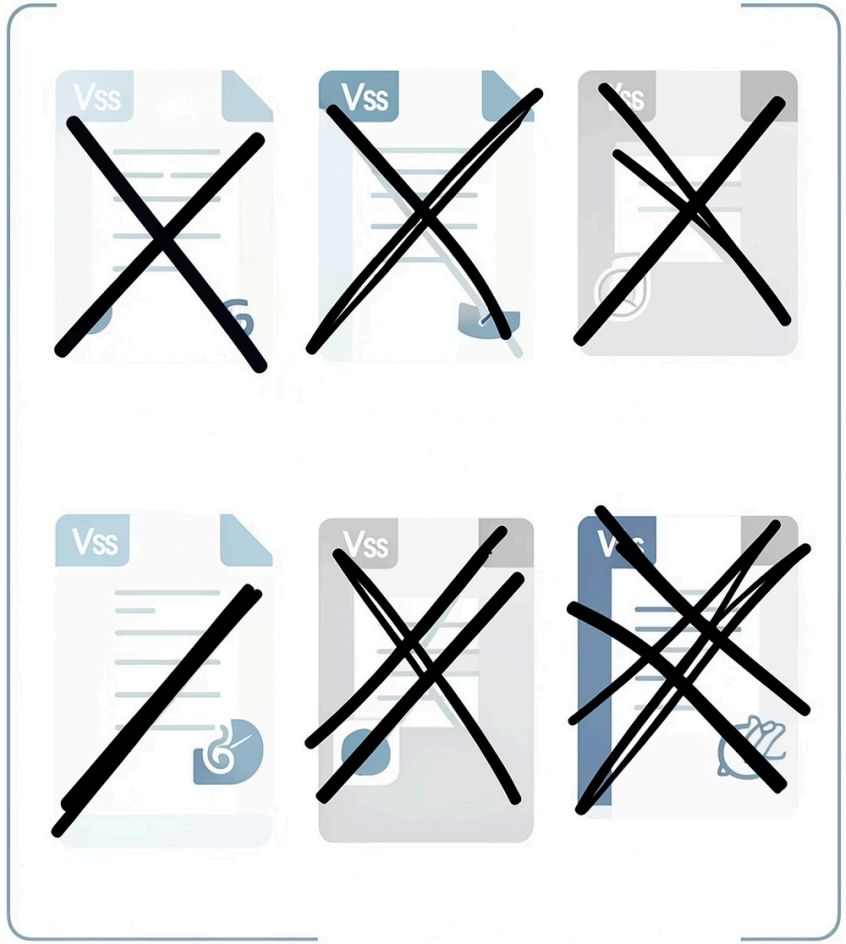

Versioning Nightmare

The Conflict

Latest Express installed locally, but server running Node v14

The Error

Cannot find module 'node:events'

The Solution

Downgraded dependencies to match environment. It's not about newest code—it's about code that *runs*.

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The Git-Bridge Workflow

01

Local Development

VS Code on Mac for speed and formatting

02

Remote Sync

GitHub as single source of truth

03

Deployment

Simple git pull on server

Why this works: A terminal is a terrible place to write complex code. Use the right tool for development, then bridge to production.

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The Vault UI

Dynamic Routing

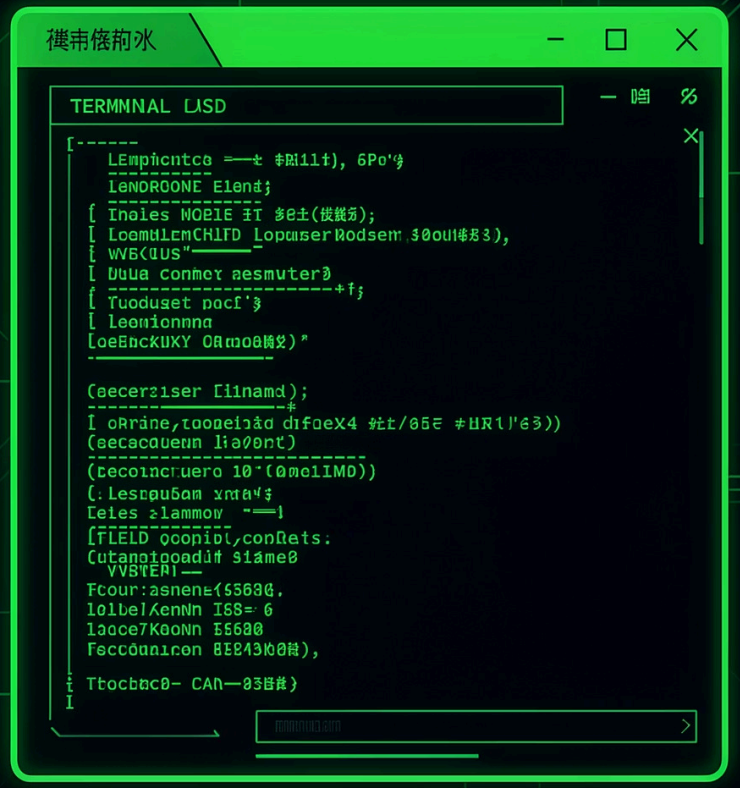
URL query parameters (?file=filename) for file selection

Security Filter

Only .md and .txt files accessible—prevents system file access

Aesthetic

Neon green on black. Classic hacker interface.



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Final Status

12:05

Uptime

AM start time

0

Vulnerabilities

npm audit approved

<5s

Deployment

From VS Code to live

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Key Takeaways

Start with constraints, not tools

Understand your environment limitations before choosing architecture

Legacy compatibility matters

Latest versions don't run on restricted infrastructure

Git is your deployment pipeline

Simple git pull beats complex automation on shared hosting

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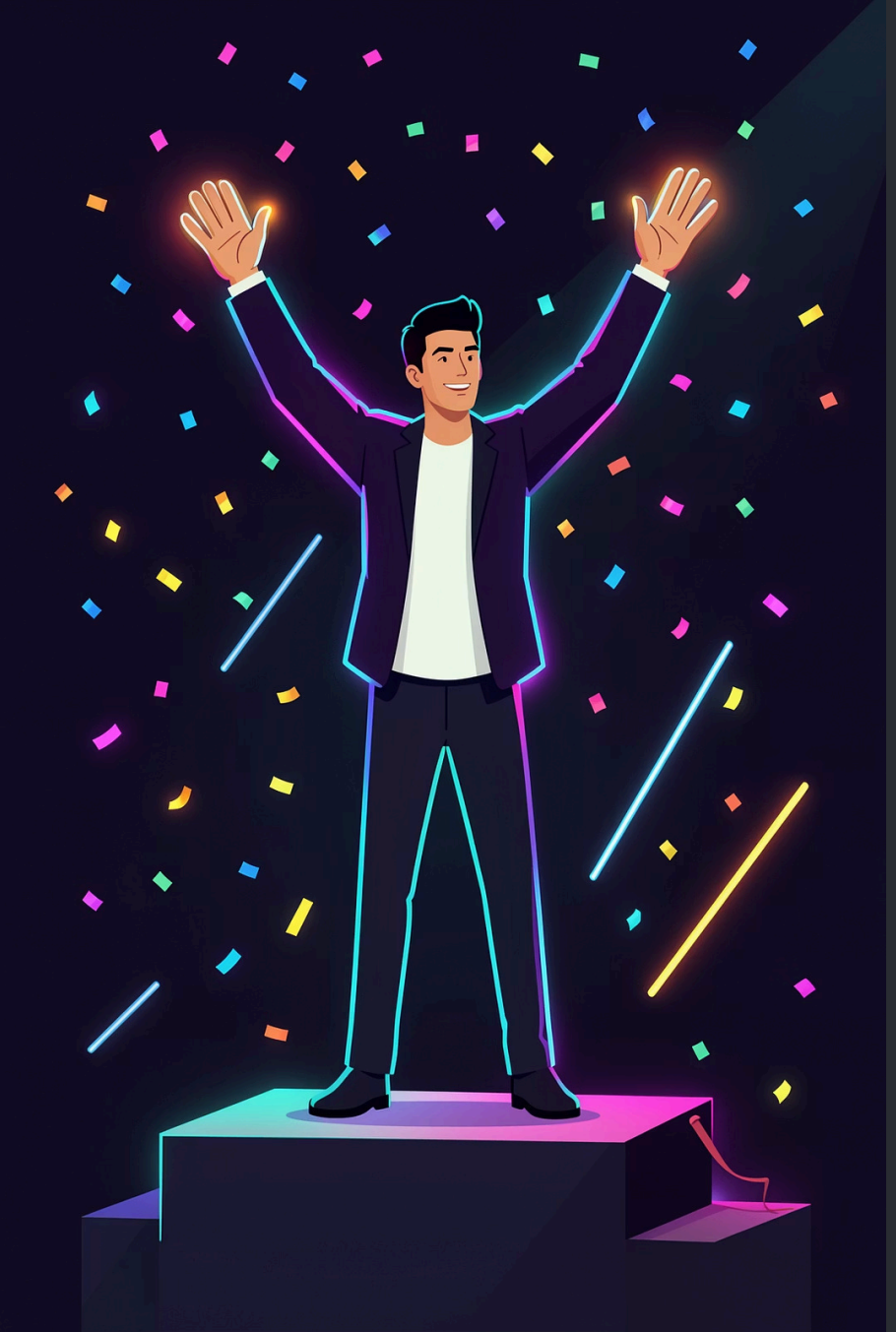

Success

Deployed live dashboard on restricted Ubuntu container using minimalist Node.js architecture, Git-based workflow, and legacy-compatible dependencies.

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